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Optimisation of a ECL Bragg Laser for Mass Entangled, Ultracold Helium Experiments

2nd Year PhB Advanced Studies Course Report
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1 Abstract

In this paper, we expand upon the previous endeavours of the ANU's He* Bose-Einstein condensation (BEC) group in developing and characterising a stable, narrow linewidth external-cavity laser (ECL) tunable over 100 nm around 1080 nm. This particular ECL has two key advantages over conventional laser diodes in that it is affordable and capable of higher performance applications such as spectroscopy. This gives ECL's a wide range of applications such as the BEC group's recently proposed precision measurements of the proton charge radius. Previous iterations of the group's ECL demonstrated promising results but were unable to reliably perform single mode lasing due to a suspected source of noise from the laser driver in the electronics control box. Our work on the ECL has indicated that the laser's control box is not responsible for any noise in the $\sim$50$-$200 mA current range that the laser operates in. The first steps in characterising the ECL produced a power$-$current curve with a slope efficiency within 0.28σ of the previous iterations. However, the threshold current remains to be 17.2σ from a previous attempt at characterising the laser. This indicates that the optics require further alignment before more sophisticated characterisations take place along with achieving single mode lasing.

2 Introduction and Motivation

Quantum electrodynamics (QED) remains to be one of the most predictive theories on the interactions of light and matter from the 20th century [1]. With high precision atomic spectroscopy offering a sensitive tool to test fundamental physics [2] at a precision now reaching the parts-per-trillion level, some disagreements between QED predictions and experimental results have arisen [1]. One discrepancy that is of particular interest is the so called 'proton radius puzzle' which was first reported in 2010 by Pohl et al [3]. The group made use of muonic hydrogen - a hydrogen atom with it's electron swapped for a muon - to show that the root mean square charge radius of the proton, a measure of the spatial distribution of the electric charge inside the proton, is $r_p = 0.84184(67)$ fm. This was quite a perplexing result given it was 5 standard deviations from the standard CODATA [4] value of $r_p = 0.8768(69)$ fm at the time. This small, yet profound, deviation from the theoretical value that can be calculated using bound-state QED [3] seems to indicate some unknown lepton-nucleus interactions beyond the Standard Model may be at play [5]. Knowledge of quantities such as the charge radius of the proton is of value in fundamental physics for determining constants such as the Rydberg constant which may be done through H and He spectroscopy [5]. Consequently, the next step is to consider the simplest multi-electron atom and investigate the muonic helium radius in comparison to the electronic helium radius to help shed light on the proton radius puzzle. This is made possible due to the progression of helium spectroscopy over the last century alongside the development of non-relativistic QED which allows for corrections up to the order of the $m\alpha^7$ term, with m the mass and $\alpha = 1/137$ the fine structure constant, being considered in the calculation of the electronic energies of helium, achieving an accuracy at the sub parts-per-billion level [6]. Another key advantage of using helium is the accessibility of its isotopes, ^{3}He and ^{4}He, allowing for a differential measurement of the two isotopic radii, thereby providing a new measurement to test QED theory with.

As of recent years, the ANU's He* Bose-Einstein condensation (BEC) facility, led by Andrew Truscott, has looked into, and proposed using its capability in readily producing ultracold clouds of both the aforementioned helium isotopes to determine the difference in the helium isotope charge radius. This in turn would offer a new perspective on the proton radius puzzle. By performing measurements in this novel way, uncertainties due to the Doppler shift, that arose from using hot atomic beams and large laser power that previous groups [7, 8] encountered will be significantly reduced leading to a more reliable determination of the isotope shift. One aspect of the experimental apparatus required for such a project is building a highly stable tunable laser system. Previous work by the group [9] led to the development and characterisation of a stable, narrow linewidth external-cavity laser (ECL) tunable over 100 nm around 1080 nm using a single-angled-facet (SAF) gain chip. The main advantage of this particular ECL is the low-cost and effectiveness as an alternative to the fibre and diode laser in this wavelength range. Whilst this laser design allows for a linewidth on the order of 10 kHz along with an absolute stability of 100 kHz, it fails to reach the sub 1 kHz level that would be required by the predictions of QED for the proposed experiment to have sufficient precision [9]. These properties are required due to helium's 1.6 MHz natural linewidth [6] and needing to suppress any long-term drift in the laser frequency. To expand upon this work, Xiwen (Andy) Qiu, a student of the ANU He* BEC group under the supervision of Sean Hodgman made some progress on developing the required ECL system and was able to construct a laser control box along with a laser housing and achieved a tunability range of 40 nm around 1080 nm [10]. The scope of this paper is to therefore build upon Andy's efforts in constructing the

laser system, with the goal of achieving single mode lasing, a property of the laser that has so far been elusive and would provide the required narrow linewidth for spectroscopy.

3 Background

Before introducing the specific design of our laser system, we will give an overview of the ECL laser. Semiconductor lasers are a class of laser that use a semiconductor as the gain medium. By supplying the medium with energy, in a process called pumping, electrons in the gain medium are excited to higher energy levels. Emitted photons then bounce back and forth in a cavity, each time going through the gain medium, further exciting more electrons. Once a population inversion occurs and provided the laser injection current is greater then its threshold, the laser starts emitting photons [11]. In general, a semiconductor laser that uses a waveguide resonant cavity extended beyond the solitary edge emitting laser is what we will consider an ECL [12]. These types of lasers have been known since at least 1964 [13] with promising progress being made in the field of semiconductor optical amplifiers as they provide physicists with a high degree of wavelength tunability [12]. There are many configurations for ECL lasers [12] but due to minimal mechanical disturbances from the low number of components, the Littrow configuration was chosen for this project [9].

3.1 The Littrow Configuration ECL

The Littrow configuration allows the diffraction grating to be positioned such that at the Littrow angle, the 1st order diffraction back-reflects from the grating into the laser diode (LD). Specular reflected light, which is the zero order diffracted light is what is then outputted by the laser through the free-space output [10]. The grating equation, with a the grating pitch, θ the angle of incidence and ϕ_m the diffraction angle for the mth order and wavelength λ is,

$$a(\sin\theta + \sin\phi_m) = m\lambda$$

and demonstrates that when we have the Littrow condition of $\phi_1 = \theta$, the equation reduces to,

$$\lambda = 2a\sin\theta \implies \frac{\mathrm{d}\lambda}{\mathrm{d}\theta} = 2a\cos\theta$$

This shows that small adjustments of the grating angle lead to the desired wavelength tunability. Importantly, this configuration allows for the diffraction grating to serve as a wavelength selective mirror by making use of the back-reflection from the grating and in turn providing a narrower linewidth compared to a bare diode laser [10]. The key features of the Littrow configuration ECL are presented in Figure 1.

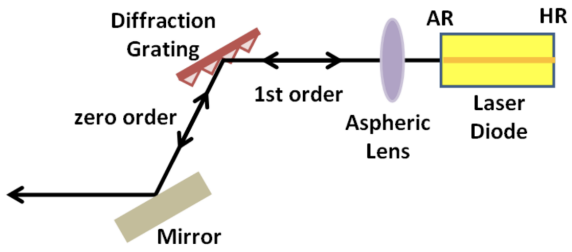


Figure 1: An example of a Littrow configuration ECL [14] where one end of the laser diode has been coated with an anti-reflective (AR) coating whilst the other end has a high-reflective (HR) coating. The diffraction grating acts as a cost effective frequency selective element [15] thereby narrowing the line width to the desired kilohertz level [10]. Here we can see the 1st order back-reflection and zero order free-space output from the Littrow ECL.

4 Laser Design

Notably, the laser diode in our ECL system contains a SAF gain chip as the gain medium. This was used to achieve wide tuning and high spectral purity [16] compared to the conventional Fabry-Pérot (FP) gain chip [10]. By construction of SAF gain chips, they contain an extended active ridge that is in the millimeter range, a whole order of magnitude greater then an FP diode, which hence admits a reduction in injection current induced frequency noise at the cost of a modulation bandwith of the ECL decreasing [9]. Another significant advantage of SAF over FP gain chips is that in an ECL configuration the stability of the intrinsic output beam pointing from the normal facet allows a stable operation that is invariant to adjustments of the Littrow angle [9]. To collimate the free-space output of the gain chip a 0.5 NA lens with 8 mm focal length at 1083 nm was implemented [9]. The custom made mount for the lens provided various translational degrees of freedom to fully optimise performance of the laser. In particular, adjustments along the Z axis up to $\sim 50\mu$m between the lens and the gain chip's front facet are possible via the finely pitched thread on the lens tube. However, variations in the XY plane are limited due to the design but are not critical for alignment [9].

As indicated in Figure 1, after the light passes through the collimation optics it diffracts off a blazed diffraction grating at 1200 lines/mm with an efficiency of 80% at 1083 nm. This was used to give control over the wavelength that

forms the external cavity [9]. A piezoelectric transducer (PZT) chip was used to provide a continuous and moderate bandwith control (∼1 kHz) of the laser frequency [10]. This was installed in between the grating and the mirror mount, as seen in Figure 2. A key consideration in the design of the laser was achieving total control over the temperature in the laser, both in the SAF gain chip and the laser housing. As such, two separate thermoelectric coolers (TECs) and thermistors were used to allow the desired temperature control. To vary the temperature of the whole laser housing, a TEC connected to a Peltier device was installed under the laser block whilst the other TEC directly controlled the temperature of the gain chip [9]. The aforementioned laser housing which contained all the critical components of the laser - the SAF gain chip, collimation optics and diffraction grating - was constructed from a single aluminium block which attenuated any background environmental noise, mainly in the form of vibrations [9]. To further limit sources of noise, the laser was enclosed in an acoustically insulated supported by viscoelastic dampers, which prevented any vibrations from the optics table from interfering with the laser.

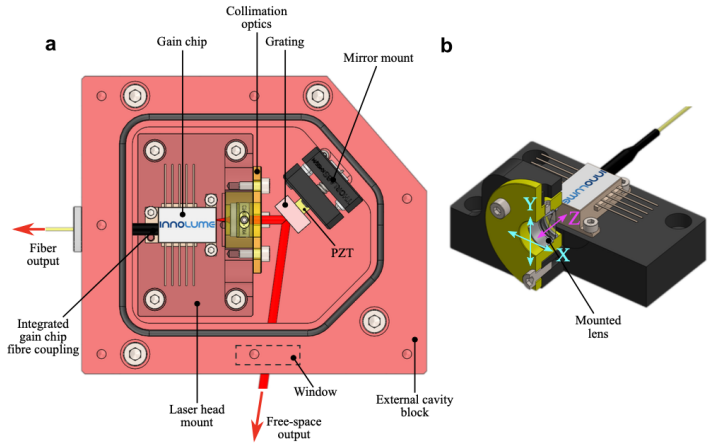


Figure 2: A schematic showing the exact design of the particular ECL laser that was under use [9]. (a) The key components that are required for the Littrow configuration as seen in Figure 1 are shown. The zero order diffraction is shown to be aligned such that it is outputted into free space, whilst the 1st order diffraction is back reflected into the fiber output. Not shown in the schematic is the Peltier device that is placed under the laser housing. Similarly, the various thermistors and TEC for the SAF gain chip are not shown. (b) A close up of the collimation optics and SAF gain chip. This shows the various degrees of freedom in the XY and Z direction that is essential for laser alignment.

5 Electronic Control Box Design

A key focus of this project was diagnosing, and then resolving suspected issues surrounding the source of the noise in our electronics control box. The purpose of the box is to house various electronic devices to allow users of the laser to vary and monitor the injection current, the temperature of the laser housing and SAF gain chip through an LED display. We used a Koheron DRV200 low noise laser diode driver to provide the injection current for the laser. A TEC-8A-24V-PID-HC was implemented to control the temperature of the laser housing between 0−50 C with an input voltage in the range 5−24 V and maximum current of 8 A. A TEC100L linear temperature controller allowed us to vary the temperature of the SAF gain chip directly from 10−40 C with a 24 hour stability of 0.005 C.

To be able to communicate with these components, we also required two more circuit elements: an Arduino Uno and an ADS1115 analog to digital converter (ADC). Through these components, we were able to convert the analog set-point and actual temperature of the gain chip along with the injection current to a digital reading from which we used an Arduino code to display these values on an LED display located on the front of the control box. This was critical in facilitating us knowing the exact temperature and current setting that the laser was operating under. Of the various components, it was expected that low noise laser diode driver, which was specified to drive a current between 0−200 mA, was responsible for the lack of single mode lasing capabilities. This is mainly because only the laser driver directly impacts the output power of the laser and would as such prevent the desired single mode lasing from occurring. As such, we carried out a rigorous testing procedure on this component by isolating it from the control box circuit and measured the peak-to-peak current over the full range of output current by measuring the potential difference over a resistor (of known resistance) and LED. This indicated that the noise was 60^{+80}_{-60} $\mathrm{nA_{rms}}$ which was within the 255 $\mathrm{nA_{rms}}$ noise tolerance, as specified by the manufacturer, for all currents apart from the 30−35 mA range in which severe noise spikes occurred as seen in Figure 3. The exact cause of the spikes could not be determined as changing the resistor in the setup did not alter the range in which the spikes occurred nor reduce the peak to peak voltage measured on the oscilloscope. The only change that seemed to provide some reduction in the spikes was removing the LED from the circuit, and just measuring the potential difference over the resistor implying that the LED was the source of some of the noise in this current range. Given that beyond the 30−35 mA current range, the peak to peak voltage reading on the oscilloscope was at most 1 mV, with a background reading measured to be 0.880(2) mV it can be concluded that the 10 mV spikes are indeed statistically significant. However, as we will later show that the threshold current of

the laser is at least 50 mA, the noise spikes being at 30−35 mA pose no negative implications for the laser's operation.

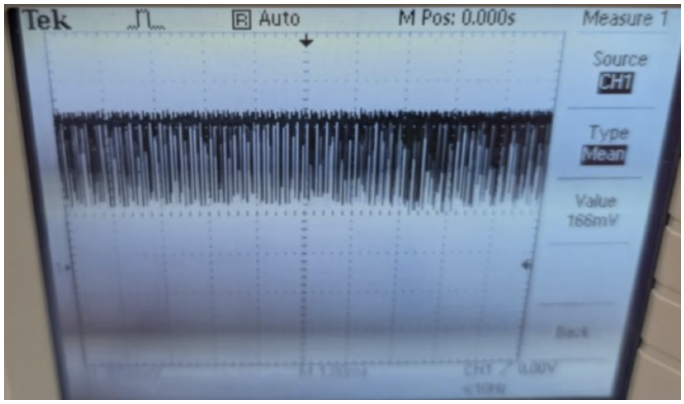


Figure 3: The distinct noise spikes that occurred in the 30−35 mA current range when testing the DRV200 laser driver. Each dotted box on the oscilloscope display represents 1 ms along the horizontal direction and 5 mV in the vertical direction.

The two TECs were not able to be tested in a similar fashion to the laser driver as they require a direct connection to the various thermistors in the laser housing and SAF gain chip and so could only be tested once the box was connected to the laser. A thermocouple was connected just below the mirror mount in Figure 2a which gave a reading on the temperature of the housing. The TEC-PID-HC has the capability to both cool and heat the housing by reversing the polarity, and hence direction of the current, in the wires that connected to the Peltier device. The TEC-PID-HC has no connection to the ADC as there was no analog output, instead it has a USB to Transitor-Transitor Logic (TTL) converter meaning that further programming using a software called PuTTY was required. However, due to technical difficulties and incompability this was later deemed unusable and so an alternate method was required. Consequently, the thermocouple was employed to calibrate the TEC-PID-HC and monitor any subsequent temperature fluctuations around the set point of 22.3 C. This particular temperature was chosen due to it being the average lab temperature.

As the TEC100L had an analog output, this was sent through the ADC to give a digital display of both the set point and actual temperature of the SAF gain chip. To confirm the validity of the displayed temperatures, the resistance of the thermistor controlling the temperature was measured and confirmed to correspond to the exact displayed temperature by a temperature-resistance curve provided by the manufacturer. Although the laser driver was confirmed to be within the manufacturer's specified noise tolerance, the current on the LED display was marginally miscalibrated to what was read off the oscilloscope so we accounted for this using the following (Figure 4) calibration curve and implemented this into the Arduino code. The rest of the Arduino code followed the structure of taking an input voltage from the ADC, then used an equation provided on the specification sheet of each component to convert the voltage into a current, for the laser driver, and a temperature, for the TECs.

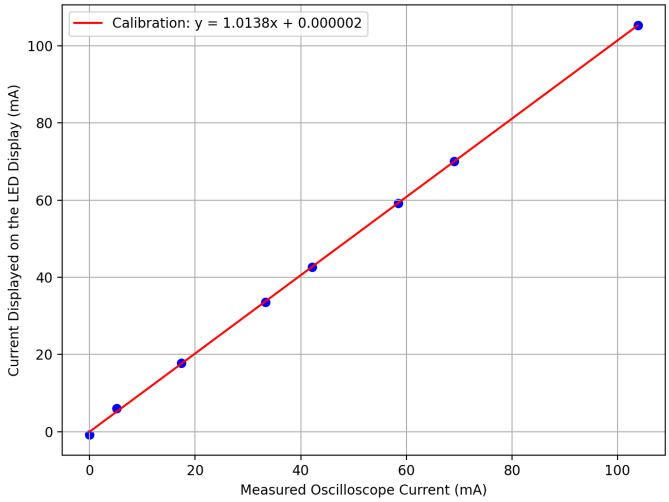


Figure 4: Calibration Curve for the Measured Current to the Displayed Current. This facilitated an accurate current reading on the control box's LED display. Note that error bars are considered, however are not visible.

5.1 Control Box Wiring

The main connections within our control box and how they interact with each other is now discussed. Figure 5 provides a circuit schematic of the main connections that were necessary for the laser to function as desired.

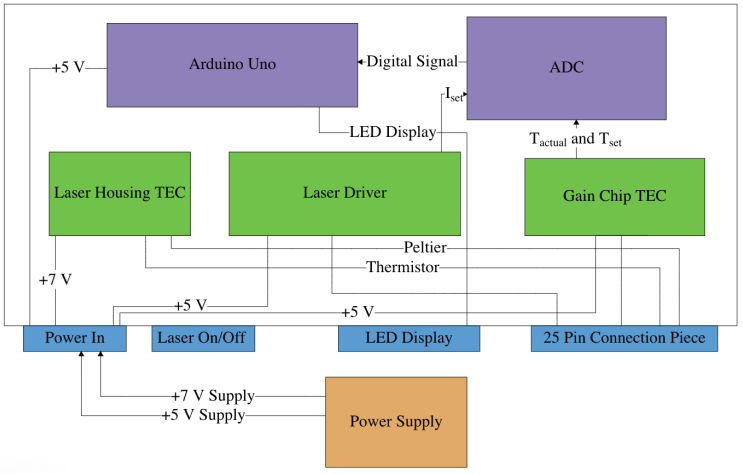


Figure 5: A circuit diagram showing the key components and wiring in the final version of the electronic control box. Note that this is not a full description of the wiring; see the Appendix for a complete circuit diagram.

Importantly, there are two power inputs - one for the TEC-PID-HC and the other for the remaining components. The main reason for this is that the TEC-PID-HC is specified to run on 7 V and can draw up to 3 A whilst either heating or cooling and then settles down to around 0.1 A when maintaining the set temperature. Conversely, the rest of the components can only operate at a maximum of 5 V and as such were powered via a separate power input. For the laser to emit photons, both the power inputs must be operational. As previously mentioned, the TEC-PID-HC does not support an analog output for temperature monitoring, so it did not connect to the Arduino and instead had a direct connection with the 25 pin connection piece, operating the Peltier device placed under the housing along with a thermistor. Both the laser driver and TEC100L connected to the Arduino via the ADC; the values were shown on the LED display and could be switched between the three options - actual gain chip temp, set-point temp, injection current - with a switch positioned adjacent to the display. To complete the circuit, both the TECs and laser driver connected to the 25 pin connection piece that could be directly plugged into the laser. It should also be noted that the Arduino−LED display connections contain a significantly more complex wiring then shown in Figure 5 but as this was done by Xiwen in [10] and worked successfully, it was left untouched during this project.

6 Laser Operation

With the electronic control box having undergone the testing discussed in the previous section, the principal goal of achieving single mode lasing with the ECL could be pursued. The alignment of the diffraction grating and the collimation optics followed a procedure given by one the He* BEC group's past PhD students, Dongki (David) Shin's, thesis [17] to optimise the power through the fibre output. As the collimation optics only require a once in a life time alignment [17], we only focused on back-coupling the 1st order diffraction back into the gain chip to maximise the fibre power output. This was done by placing a sheet of paper in front of the free-space output of the laser (see Figure 2a) and using a hand-held infrared viewer to observe its output. By varying the grating angle (access to which is located on the diagonal edge of Figure 2a) a variation in the intensity of the free-space output is caused. This is done until the output noticeably increases in intensity which may be measured by attaching an optical power meter to an optics stand and then monitoring the output on the display of a power meter. Once the free-space output achieves a maximum, this implies that the 1st order diffraction is being back-reflected into the gain chip and finer optimisation of the power can be achieved by monitoring the fibre output directly with the power meter and applying fine adjustments to the mirror mount. The nature of the maximum should also be considered, as it may just be a local maximum but due to there only being one angle in which the Littrow condition is met then this guarantees any other maxima would give an equivalent alignment. It should be noted that even a very slight adjustment of the mount, even inserting and removing the screwdriver to change the set-point, was enough to noticeably change the fibre output and so extreme care is required when aligning the ECL. However, if fine adjustments to the grating angle need to be made, this can easily be accomplished through the use of the PZT.

6.1 Power-Current Response

By varying the injection current, one can find the threshold current of the ECL configuration - the current at which the laser starts lasing. For our initial alignment, a power−current plot revealed the threshold current to be 76.3(4) mA, considerably higher then previous attempts that achieved 59.35 mA [10] and 50(1) mA [9]. As such, further optimisation of the laser's alignment following David's alignment procedure was carried out. Through this process, the threshold current was further reduced by 10 mA to a more reasonable, yet still non-optimal, 64.5(3) mA allowing us to produce the following power−current curve (Figure 6).

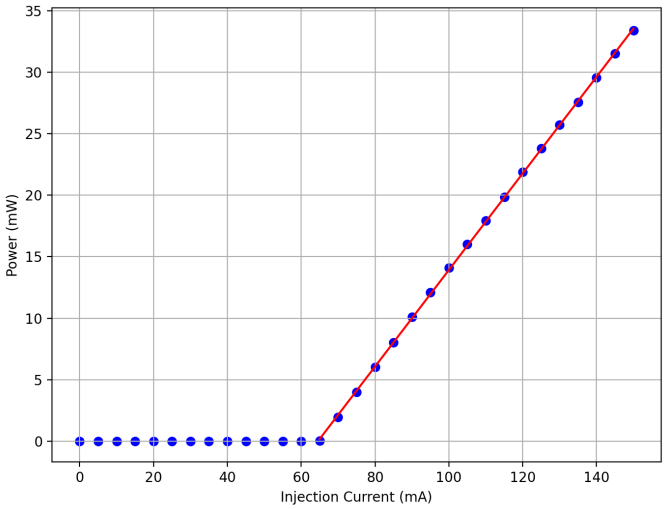


Figure 6: The power−current curve after optimisation of the laser alignment. The power output (y axis) is at the fibre output, as a function of the injection current (x axis) which was at 1083.33(1) nm. The threshold curent (x-intercept) was determined to be 64.5(3) mA with a slope efficiency of 0.39(1) mW/mA. The data gives $R^2 = 0.99$ indicative of a strong correlation. Note that error bars are considered, however are not visible.

Whilst the threshold current from both previous setups [10, 17] can be compared as this is a function of the alignment, the slope efficiency is a function of the gain chip and as a different one is used between the two papers, only Xiwen's [10] result can be compared to. Indeed, we report a value of 0.39(1) mW/mA which can be checked to be within 0.28σ of Xiwen's value of 0.362 mW/mA which was reported without an uncertainty. This indicates that our efficiency values are consistent with each other implying that the gain chip is operating as expected. However, some concern is raised when comparing our threshold current to Xiwen's value of 59.35 mA given there is a 17.2σ difference, suggesting a major statistical disagreement in the values. As the alignment procedure was unable to reduce the threshold current beyond 64.5(3) mA, this perhaps suggests that the collimation optics are the limiting factor in optimising the threshold current. To confirm this, the procedure given by [17] to realign them would have to be carried out and so the mirror mount (in Figure 2a) would have to be removed giving access to the degrees of freedom of the optics in Figure 2b. From there, the same process of obtaining zero order back reflection would have to be completed to once again achieve the Littrow configuration alignment. Due to time constraints on the project, this was not done but provides a starting point for future work. It should also be noted that although we only test currents in the range $0-150$ mA, this does not limit the scope of the laser applications as the fibre output can be connected to an optical fibre amplifier. In this case, only a quarter of the SAF gain chips maximum current is required and as such extends the operational lifetime of the laser [10].

7 Conclusions and Future Outlook

This paper has presented the work conducted in assessing suspected noisy components in the ECL electronics control box, followed by the initial characterization of the ECL. Although several major characterisations of the laser are missing, such as a demonstration of single mode lasing or a spectrum scan of the 2^3S-2^3P transition of ^4He as completed by [8] (which would be a long term objectiv), we have been able to confirm that the control laser box has an acceptable noise level, below that specified by the manufacturer. Further, an initial verification of the slope efficiency of the laser shows that we are within uncertainty to previous work performed by Xiwen [10]. However, the threshold current being beyond uncertainty of Xiwen's paper [10] indicates that further alignment of the collimation optics is required before further characterisation of the laser can take place. In any case, this paper once again supports the evidence provided by [9, 10, 17] of the ECL being a cheap and cost effective replacement of lasers that were previously designed for similar applications. In particular, the theory behind the Littrow ECL leads it to be a fairly simple laser to operate. One of the main motivators for this experiment was to reduce the laser's line width to the sub 1 kHz level that is required by the predictions of QED for spectrum scans. This still remains an open problem, given a lack of time for us to perform a heterodyne detection of two similarly constructed lasers by a fast photodetector [9]; hence, providing us with the spectral linewidth of the laser. It has also not been verified if the noise in the control box was the reason for single mode lasing being elusive, as we have not been able to carry out the single mode lasing procedure due to time constraints. This can be checked once further optimisation of the alignment of the ECL's optics is conducted.

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